Problem Set 3

Your Name

2026-10-31

https://github.com/LucyJoJohnson/EC422problemset3Johnson

##

```
# Read the CSV file + Inspect the Date Column
library(tidyverse)
retail_raw <- read_csv("data/retail_sales_raw.csv")

head(retail_raw)
```

```
# A tibble: 6 x 3
  observation_date  RSAFS RSAFSNA
  <date>            <dbl>   <dbl>
1 1992-01-01       159177  142051
2 1992-02-01       159189  142498
3 1992-03-01       158647  154439
4 1992-04-01       159921  158430
5 1992-05-01       160471  164788
6 1992-06-01       161205  163417
```

```
retail_raw <- read_csv(
  "data/retail_sales_raw.csv",
  na = ".",
  col_types = cols(
    observation_date = col_date(format = "%Y-%m-%d"),
    RSAFS = col_double(),
    RSAFSNA = col_double()
  )
)
```

```r
# 2.3 Reshape your data to long format
retail_raw_long <- retail_raw |>
  select(observation_date, RSAFS, RSAFSNA) |>
  pivot_longer(cols = c(RSAFS, RSAFSNA), names_to = "series_id", values_to = "sales")
head(retail_raw_long)
```

```
# A tibble: 6 x 3
  observation_date series_id  sales
  <date>           <chr>      <dbl>
1 1992-01-01       RSAFS     159177
2 1992-01-01       RSAFSNA   142051
3 1992-02-01       RSAFS     159189
4 1992-02-01       RSAFSNA   142498
5 1992-03-01       RSAFS     158647
6 1992-03-01       RSAFSNA   154439
```

```r
# 2.4 Build your tsibble
library(fpp3)

retail_ts <- retail_raw_long |>
  mutate(Month = yearmonth(observation_date)) |>
  as_tsibble(index = Month, key = series_id)
head(retail_ts)
```

```
# A tsibble: 6 x 4 [1M]
# Key:       series_id [1]
  observation_date series_id  sales    Month
  <date>           <chr>      <dbl>    <mth>
1 1992-01-01       RSAFS     159177 1992 Jan
2 1992-02-01       RSAFS     159189 1992 Feb
3 1992-03-01       RSAFS     158647 1992 Mar
4 1992-04-01       RSAFS     159921 1992 Apr
5 1992-05-01       RSAFS     160471 1992 May
6 1992-06-01       RSAFS     161205 1992 Jun
```

```r
has_gaps(retail_ts)
```

```
# A tibble: 2 x 2
  series_id .gaps
  <chr>     <lgl>
```

```
1 RSAFS     FALSE
2 RSAFSNA   FALSE
```

```
# 2.5 Canadian Gas Production
head(canadian_gas)
```

```
# A tsibble: 6 x 2 [1M]
     Month Volume
     <mth>  <dbl>
1 1960 Jan   1.43
2 1960 Feb   1.31
3 1960 Mar   1.40
4 1960 Apr   1.17
5 1960 May   1.12
6 1960 Jun   1.01
```

```
# canadian_gas has less columns than retail_ts, and only has one series
key_vars(canadian_gas)
```

```
character(0)
```

```
index_var(canadian_gas)
```

```
[1] "Month"
```

```
can_gas <- canadian_gas |>
  select(Month, Volume)
has_gaps(can_gas)
```

```
# A tibble: 1 x 1
  .gaps
  <lgl>
1 FALSE
```

```
head(can_gas)
```

```
# A tsibble: 6 x 2 [1M]
     Month Volume
     <mth>  <dbl>
1 1960 Jan   1.43
2 1960 Feb   1.31
3 1960 Mar   1.40
4 1960 Apr   1.17
5 1960 May   1.12
6 1960 Jun   1.01
```

```
# 2.6 Working with Dates in a tsibble
can_gas_recent <- can_gas |>
  filter(Month >= yearmonth("1990 Jan"))

head(can_gas_recent)
```

```
# A tsibble: 6 x 2 [1M]
     Month Volume
     <mth>  <dbl>
1 1990 Jan   12.5
2 1990 Feb   11.2
3 1990 Mar   11.6
4 1990 Apr   11.1
5 1990 May   11.0
6 1990 Jun   10.1
```

```
can_gas_recent <- can_gas_recent |>
  mutate(year    = year(Month),
         month   = month(Month, label = TRUE),
         quarter = quarter(Month))
```

```
can_gas_recent |>
  as_tibble() |>
  group_by(month) |>
  summarize(avg_volume = mean(Volume))
```

```
# A tibble: 12 x 2
   month avg_volume
   <ord>      <dbl>
 1 Jan         17.0
 2 Feb         15.3
```

```
 3 Mar          16.3
 4 Apr          15.7
 5 May          15.7
 6 Jun          14.9
 7 Jul          15.6
 8 Aug          15.6
 9 Sep          15.1
10 Oct          16.2
11 Nov          16.2
12 Dec          17.1
```

```
can_gas_recent |>
  mutate(next_month           = Month + 1,
         same_month_last_year = Month - 12) |>
  head()
```

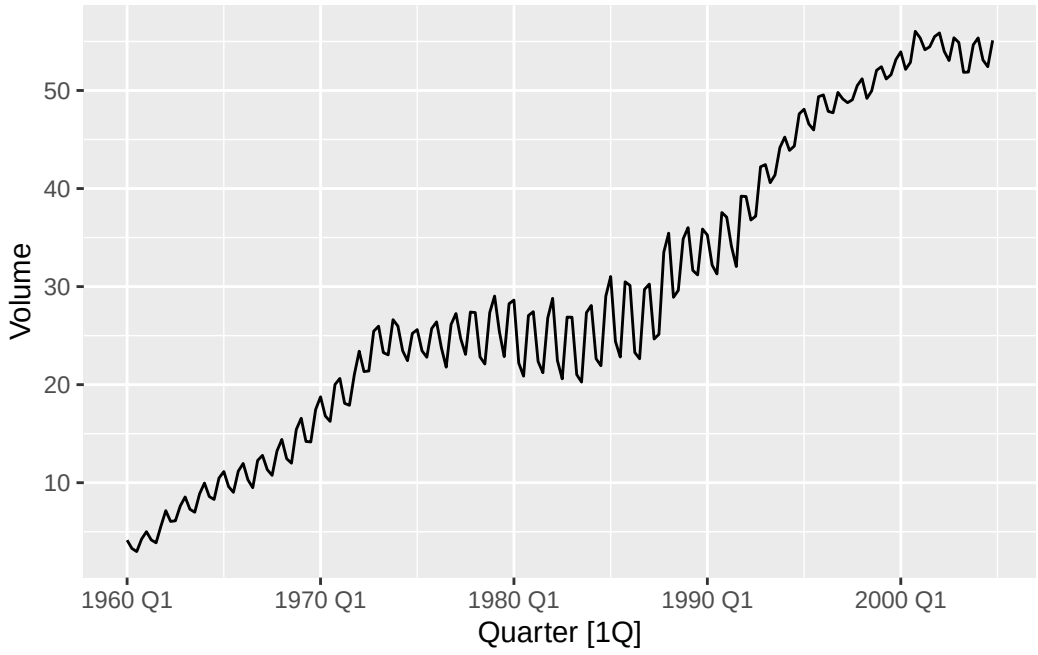
```
# A tsibble: 6 x 7 [1M]
     Month Volume  year month quarter next_month same_month_last_year
     <mth>  <dbl> <dbl> <ord>   <int>      <mth>                <mth>
1 1990 Jan   12.5  1990 Jan         1   1990 Feb             1989 Jan
2 1990 Feb   11.2  1990 Feb         1   1990 Mar             1989 Feb
3 1990 Mar   11.6  1990 Mar         1   1990 Apr             1989 Mar
4 1990 Apr   11.1  1990 Apr         2   1990 May             1989 Apr
5 1990 May   11.0  1990 May         2   1990 Jun             1989 May
6 1990 Jun   10.1  1990 Jun         2   1990 Jul             1989 Jun
```

```
can_gas_q <- can_gas |>
  filter(Month <= yearmonth("2004 Dec")) |>
  index_by(Quarter = yearquarter(Month)) |>
  summarize(Volume = sum(Volume))

head(can_gas_q)
```

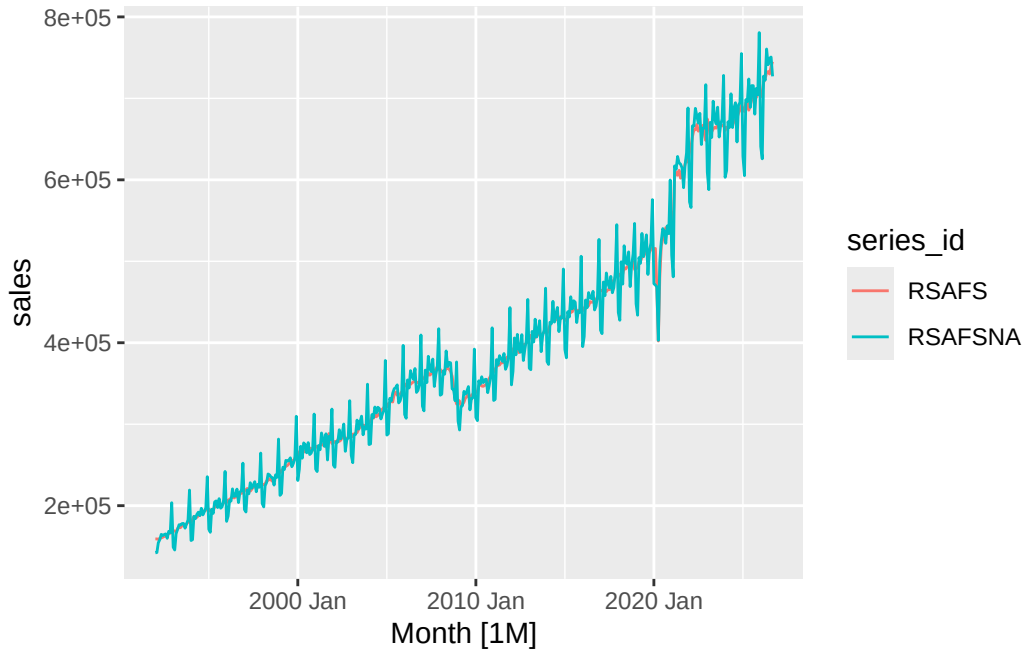
```
# A tsibble: 6 x 2 [1Q]
  Quarter Volume
    <qtr>  <dbl>
1 1960 Q1   4.14
2 1960 Q2   3.30
3 1960 Q3   2.97
4 1960 Q4   4.26
5 1961 Q1   5.01
6 1961 Q2   4.17
```

```
can_gas_q |> autoplot(Volume)
```

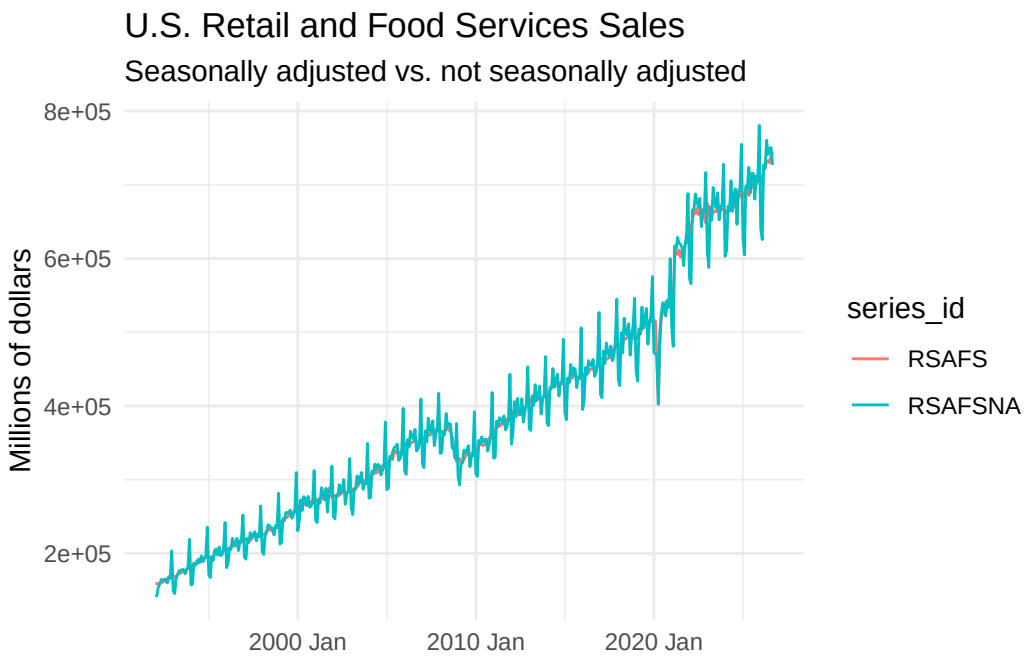



```
# We sum instead of take the mean of the monthly values of gas production because we want the
```

```
# 2.7 Plotting with Autoplot
retail_ts |> autoplot(sales)
```

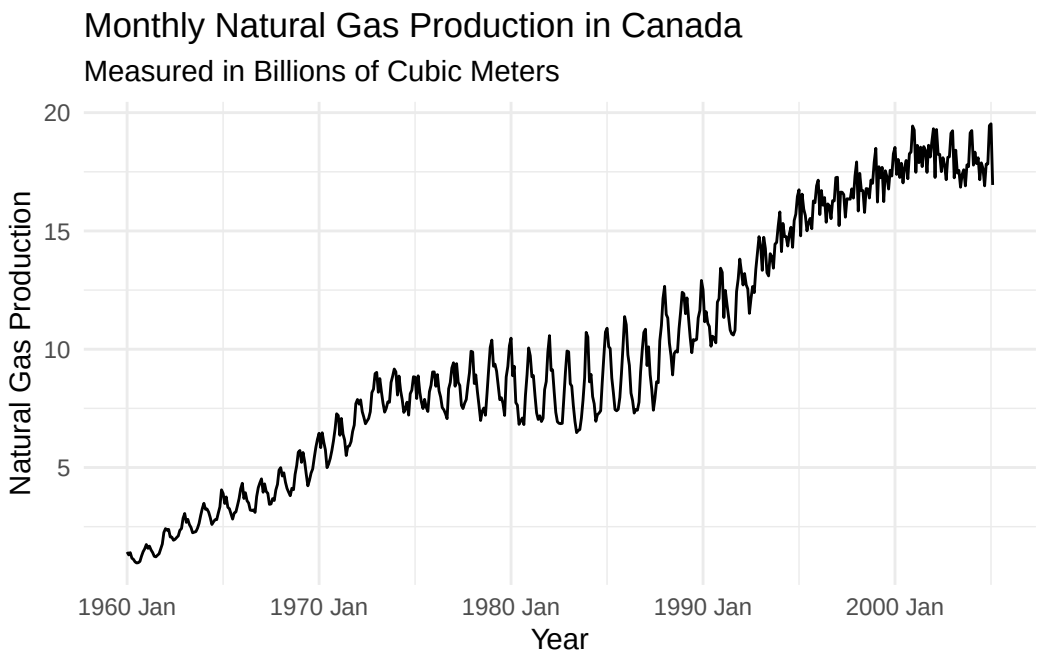



```
retail_ts |>
  autoplot(sales) +
  labs(
    title = "U.S. Retail and Food Services Sales",
    subtitle = "Seasonally adjusted vs. not seasonally adjusted",
    x = NULL, y = "Millions of dollars", color = NULL
  ) +
  theme_minimal()
```

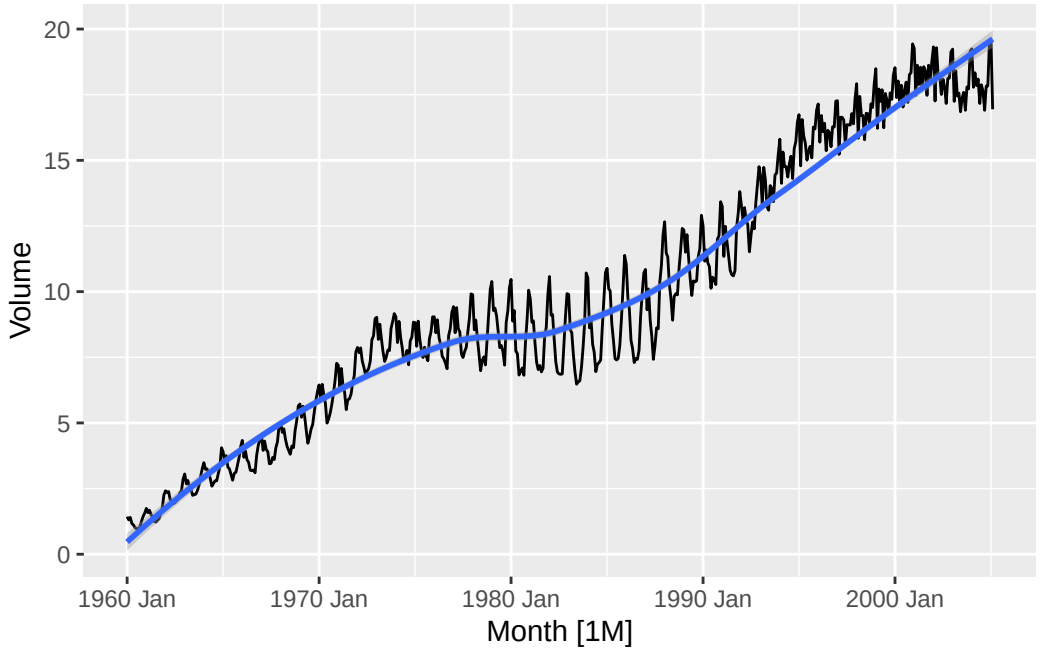



```
# The seasonal adjustment is smoothing out the line so there is less spikes and dips. The rep
```

```
library(ggplot2)
can_gas |>
  autoplot() +
  labs(
    title = "Monthly Natural Gas Production in Canada",
    subtitle = "Measured in Billions of Cubic Meters",
    x = "Year", y = "Natural Gas Production", color = NULL
  ) +
  theme_minimal()
```




```
can_gas |>
  autoplot() +
  geom_smooth()
```



```
  labs(
    title = "Monthly Natural Gas Production in Canada",
    subtitle = "Measured in Billions of Cubic Meters",
    x = "Year", y = "Natural Gas Production", color = NULL
  ) +
  theme_minimal()
```

```
NULL
```

```
# I chose geom_smooth because I wanted to see clearly how the trend went up because there are
```

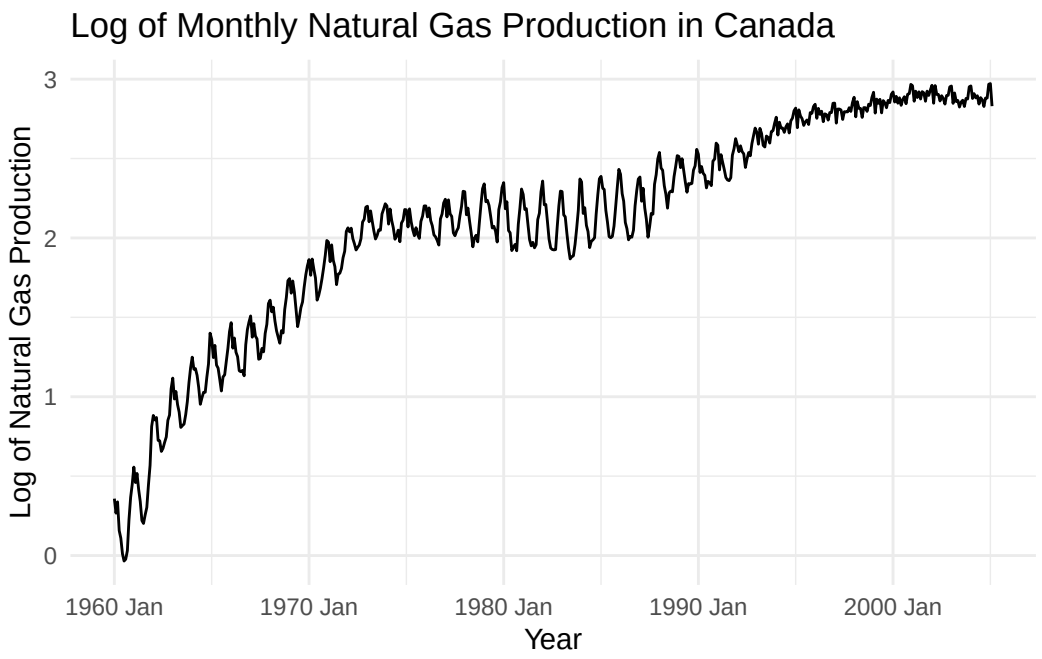
##

##

```
# 3.1 Log Transformation
# The swings get larger as the series grows, but only in the middle of the graph. Then, the s
  can_gas <- can_gas |>
    mutate(log_volume = log(Volume))
head(can_gas)
```

```
# A tsibble: 6 x 3 [1M]
     Month Volume log_volume
     <mth>  <dbl>      <dbl>
1 1960 Jan   1.43     0.358
2 1960 Feb   1.31     0.267
3 1960 Mar   1.40     0.338
4 1960 Apr   1.17     0.157
5 1960 May   1.12     0.110
6 1960 Jun   1.01     0.0112
```

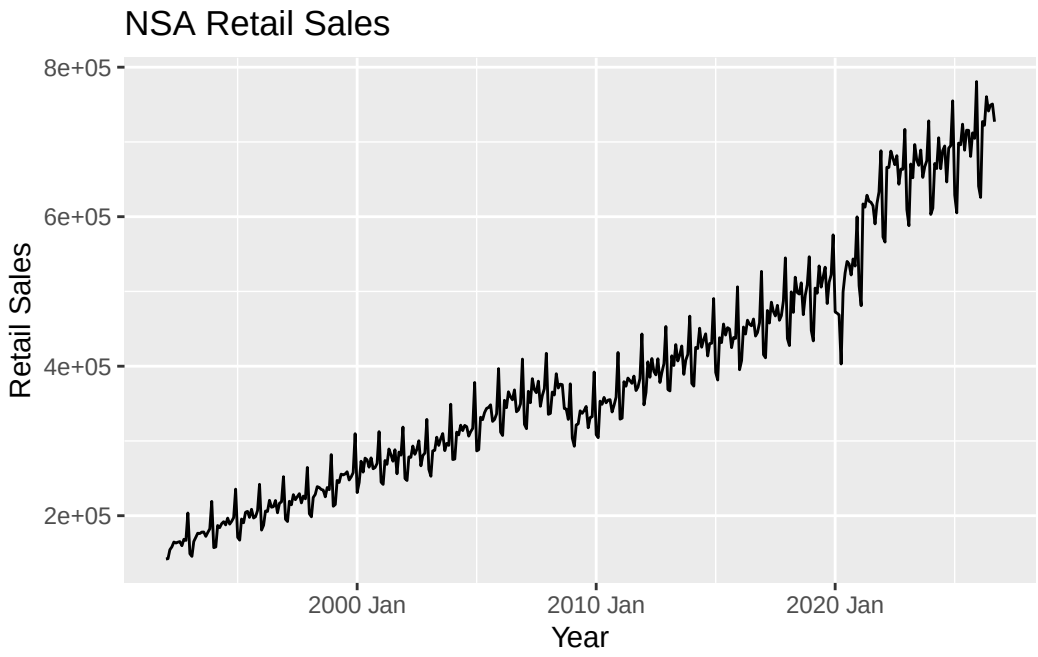
```
can_gas |>
  autoplot(log_volume) +
  labs(
    title = "Log of Monthly Natural Gas Production in Canada",
    x = "Year",
    y = "Log of Natural Gas Production"
  ) +
  theme_minimal()
```



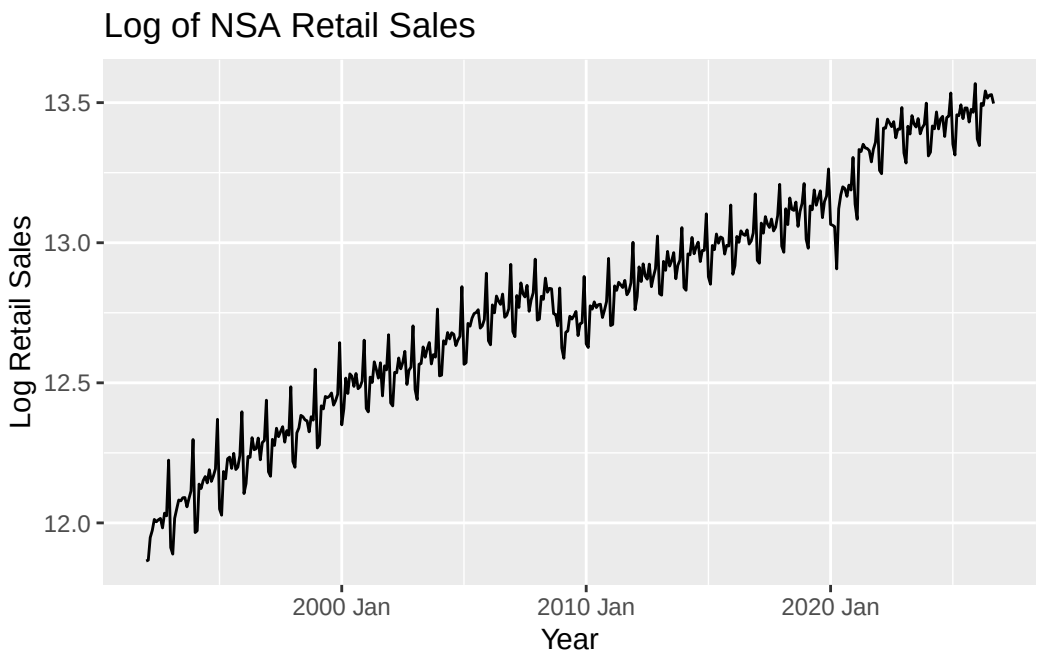

```
# The plot is still trending upward, but at an overall decreasing rate instead of increasing
```

```
retail_nsa <- retail_ts |>
  filter(series_id == "RSAFSNA")
retail_nsa <- retail_nsa |>
  mutate(log_sales = log(sales))
```

```
retail_nsa |>
  autoplot(sales) +
  labs(
    title = "NSA Retail Sales",
    x = "Year",
    y = "Retail Sales"
  )
```




```
retail_nsa |>
  autoplot(log_sales) +
  labs(
    title = "Log of NSA Retail Sales",
    x = "Year",
    y = "Log Retail Sales"
  )
```

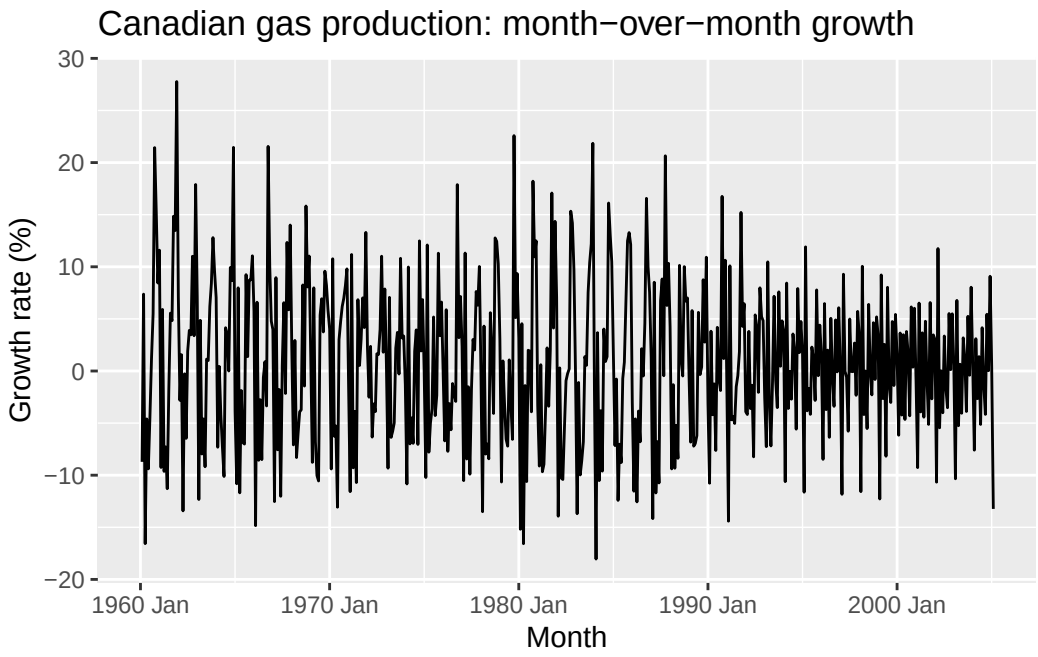



```
# I think that a log transformation for retail sales is better than for can_gas because with
```

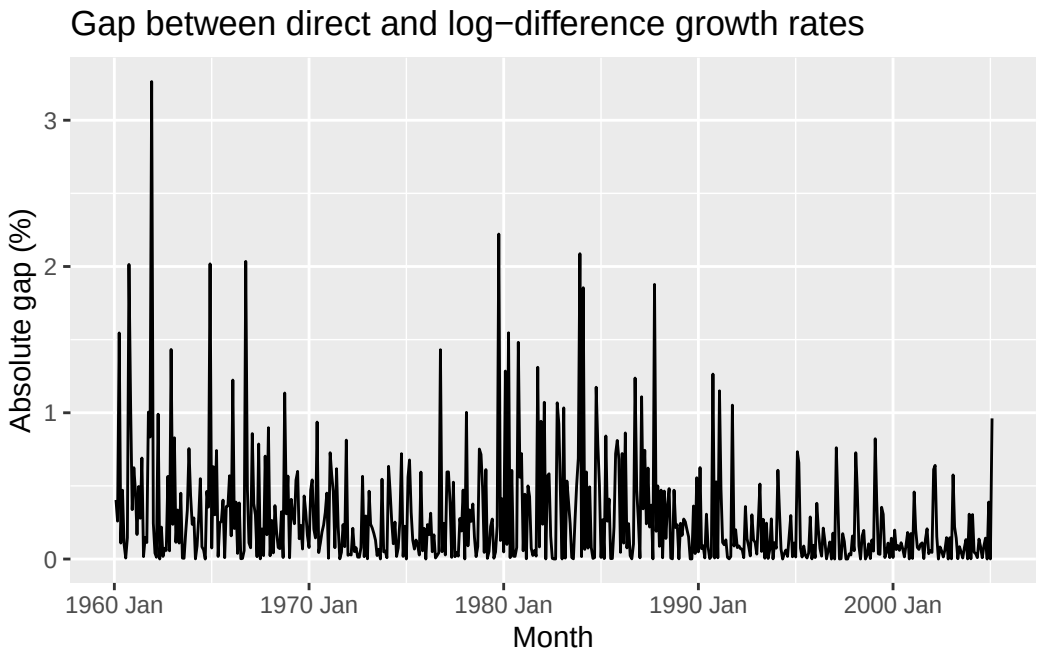
```
# 3.2 Growth Rates
can_gas <- can_gas |>
  mutate(
    growth_direct = 100 * (Volume - lag(Volume)) / lag(Volume),
    growth_log    = 100 * (log(Volume) - lag(log(Volume)))
  )
```

```
can_gas <- can_gas |>
  mutate(
    growth_direct = 100 * difference(Volume) / lag(Volume),
    growth_logdiff = 100 * difference(log_volume)
  )
```

```
can_gas |>
  autoplot(growth_direct) +
  labs(
    title = "Canadian gas production: month-over-month growth",
    x = "Month",
    y = "Growth rate (%)"
  )
```



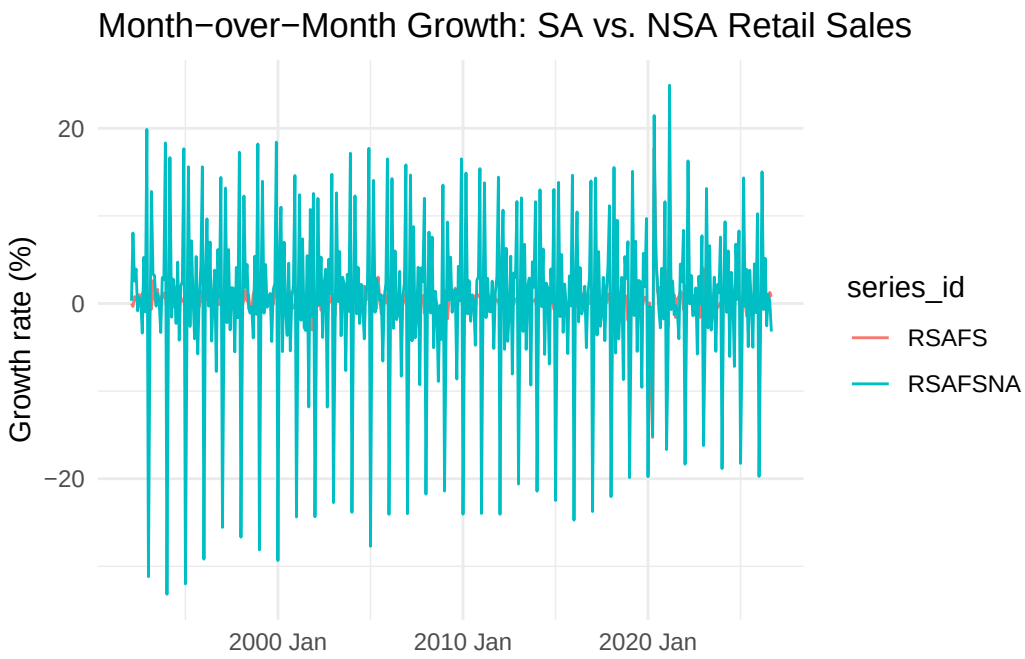

```
can_gas <- can_gas |>
  mutate(growth_gap = abs(growth_direct - growth_logdiff))
can_gas |>
  autoplot(growth_gap) +
  labs(
    title = "Gap between direct and log-difference growth rates",
    x = "Month",
    y = "Absolute gap (%)"
  )
```




```
# The log approximation is very good because the farthest difference is just above 3 percent
```

```
# 3.3 SA vs NSA Growth Rates
retail_ts <- retail_ts |>
  group_by_key() |>
  mutate(growth_mom = 100 * difference(log(sales))) |>
  ungroup()
```

```
retail_ts |>
  autoplot(growth_mom) +
  labs(
    title = "Month-over-Month Growth: SA vs. NSA Retail Sales",
    x = NULL, y = "Growth rate (%)", color = NULL
  ) +
  theme_minimal()
```

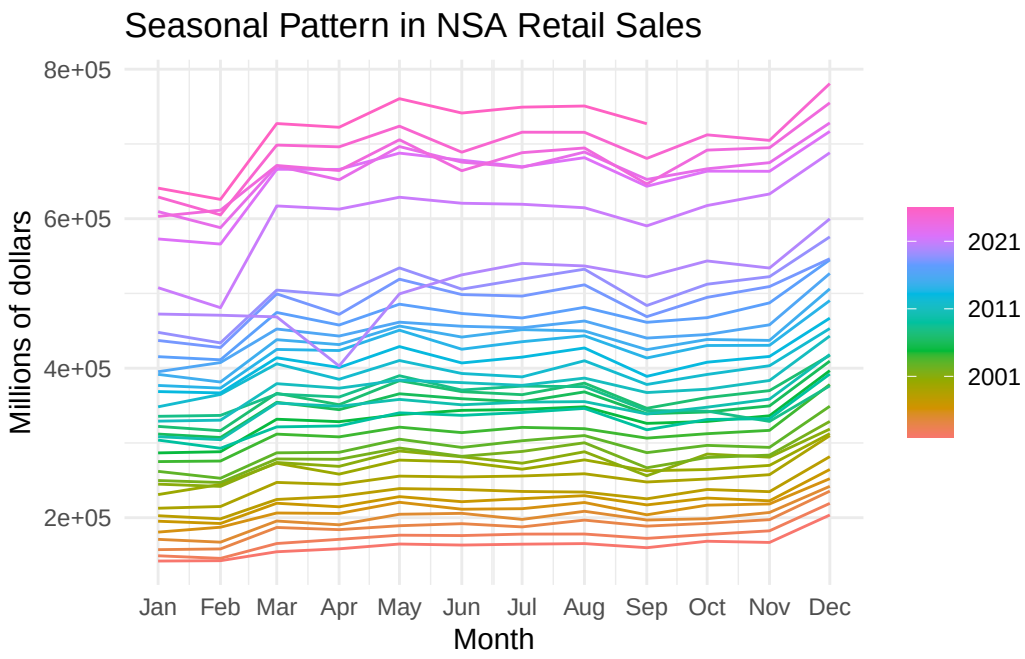



```
# The non-seasonally adjusted data has a lot more extreme spikes and dips than the seasonally
```

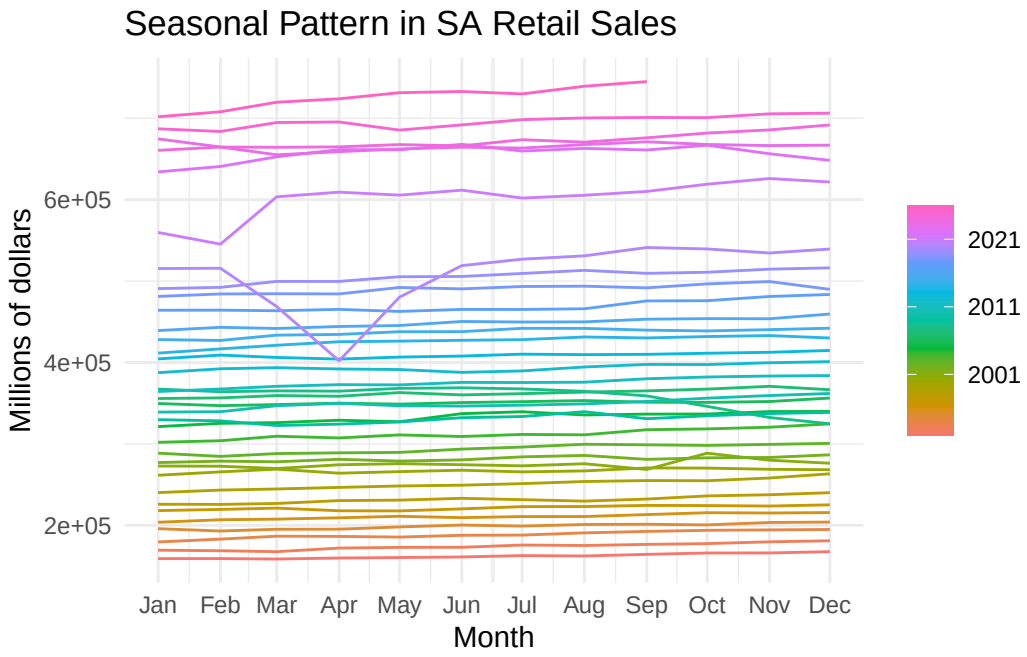
##

##

```
# 4.1 A Season Plot
retail_ts |>
  filter(series_id == "RSAFSNA") |>
  gg_season(sales) +
  labs(
    title = "Seasonal Pattern in NSA Retail Sales",
    y = "Millions of dollars"
  ) +
  theme_minimal()
```

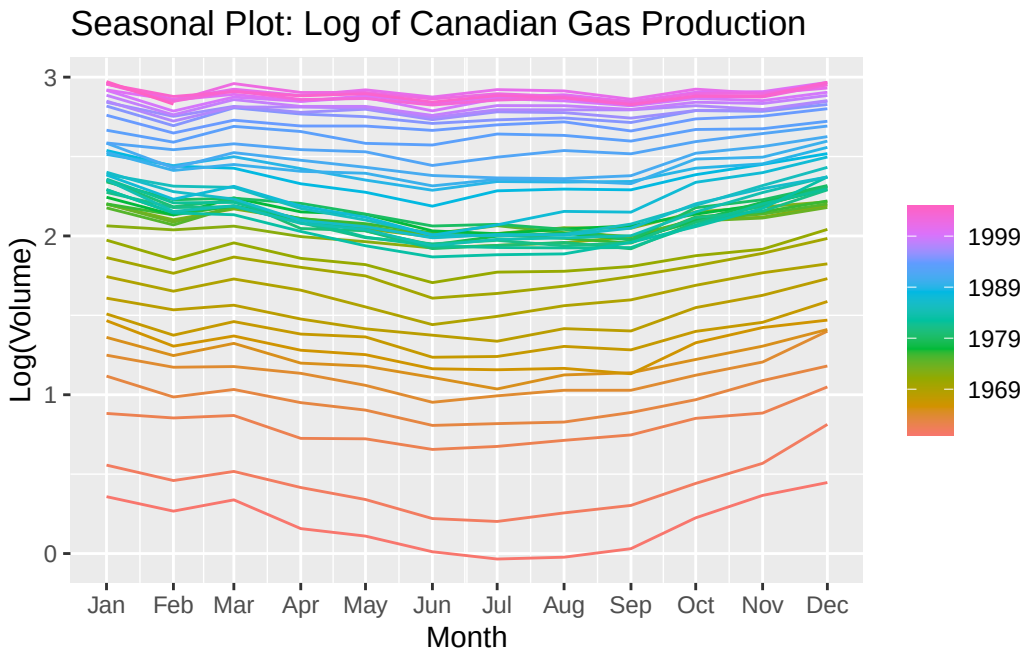



```
retail_ts |>
  filter(series_id == "RSAFS") |>
  gg_season(sales) +
  labs(
    title = "Seasonal Pattern in SA Retail Sales",
    y = "Millions of dollars"
  ) +
  theme_minimal()
```




```
# The non-seasonally adjusted graph has a rise around February and November, while the seasor
```

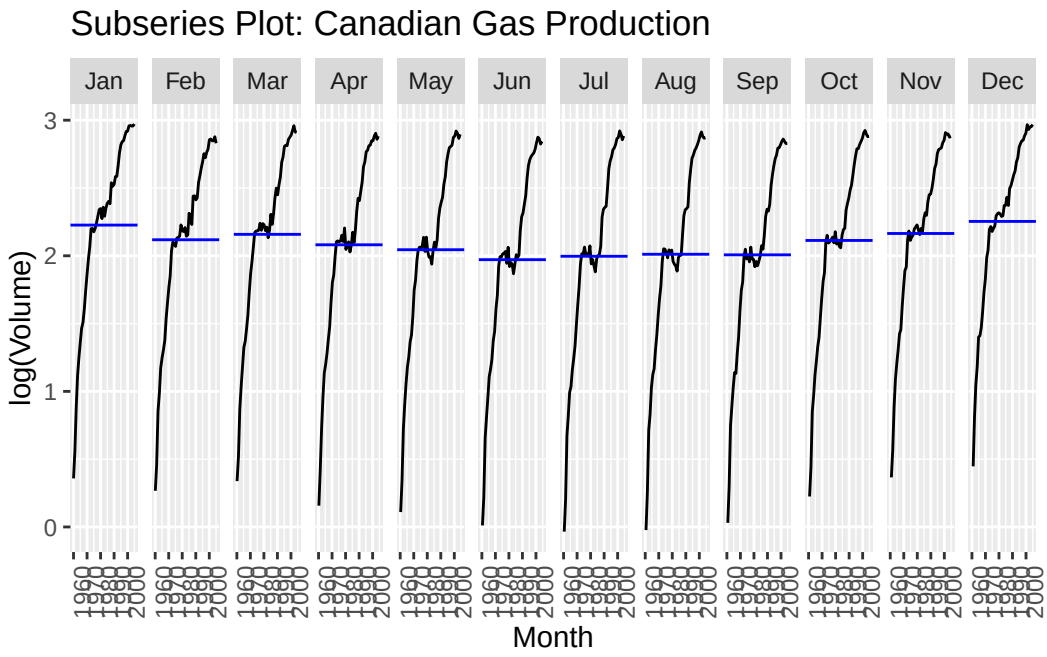
```
can_gas |>
  gg_season(log_volume) +
  labs(
    title = "Seasonal Plot: Log of Canadian Gas Production",
    x = "Month",
    y = "Log(Volume)"
  )
```




```
# I see a relatively small seasonal pattern, but there generally is a dip from March to Nover
```

```
# 4.2 A Subseries Plot

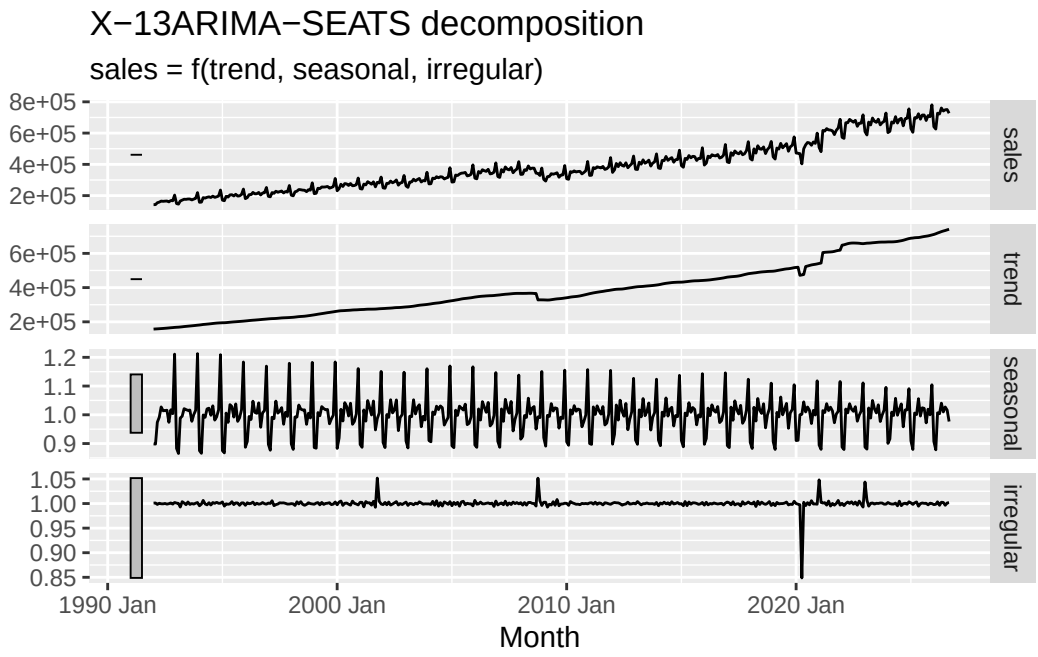
can_gas |>
  gg_subseries(log_volume) +
  labs(
    title = "Subseries Plot: Canadian Gas Production",
    y = "log(Volume)"
  )
```




```
# 4.3 Seasonally Adjusting a Series with X-13ARIMA-SEATS

retail_nsa_x13 <- retail_ts |>
  filter(series_id == "RSAFSNA") |>
  model(x13 = X_13ARIMA_SEATS(sales)) |>
  components()
```

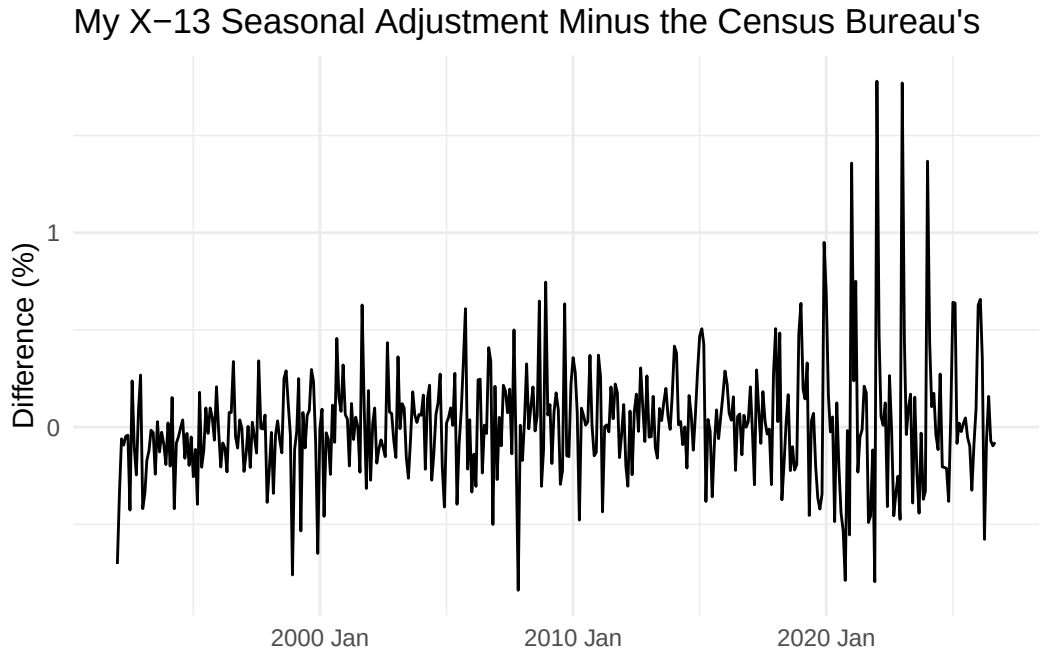
```
retail_nsa_x13 |> autoplot()
```




```
# December is when the seasonal factor is highest. In December it is 1.2 units above sales in
```

```
sa_compare <- retail_nsa_x13 |>
  as_tibble() |>
  select(Month, mine = season_adjust) |>
  left_join(
    retail_ts |>
      filter(series_id == "RSAFS") |>
      as_tibble() |>
      transmute(Month, census = sales),
    by = "Month"
  ) |>
  mutate(pct_diff = 100 * (log(mine) - log(census)))
```

```
sa_compare |>
  ggplot(aes(x = Month, y = pct_diff)) +
  geom_line() +
  labs(
    title = "My X-13 Seasonal Adjustment Minus the Census Bureau's",
    x = NULL, y = "Difference (%)"
  ) +
  theme_minimal()
```




```
# My adjustment closely matches the Census Bureau adjustment. However, the two adjustments di
```

```
# 4.4. A Simple Model - Linear Trend + Seasonal Dummies
can_gas_fit <- can_gas |>
  model(TSLM(log_volume ~ trend() + season()))

report(can_gas_fit)
```

```
Series: log_volume
Model: TSLM

Residuals:
      Min        1Q    Median        3Q       Max
-0.985307 -0.096451 -0.004525  0.131639  0.470564

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     1.150e+00  4.142e-02  27.761  < 2e-16 ***
trend()         3.973e-03  6.886e-05  57.692  < 2e-16 ***
season()year2  -1.121e-01  5.229e-02  -2.144  0.03245 *
season()year3  -5.238e-02  5.258e-02  -0.996  0.31963
season()year4  -1.334e-01  5.258e-02  -2.537  0.01146 *
```

```
season()year5  -1.738e-01  5.258e-02  -3.305  0.00101 **
season()year6  -2.512e-01  5.258e-02  -4.777 2.30e-06 ***
season()year7  -2.300e-01  5.258e-02  -4.374 1.47e-05 ***
season()year8  -2.193e-01  5.258e-02  -4.170 3.56e-05 ***
season()year9  -2.274e-01  5.258e-02  -4.325 1.83e-05 ***
season()year10 -1.253e-01  5.258e-02  -2.383  0.01753 *
season()year11 -7.781e-02  5.258e-02  -1.480  0.13956
season()year12  7.202e-03  5.258e-02   0.137  0.89111
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2508 on 529 degrees of freedom
Multiple R-squared: 0.8652, Adjusted R-squared: 0.8622
F-statistic:   283 on 12 and 529 DF, p-value: < 2.22e-16
```

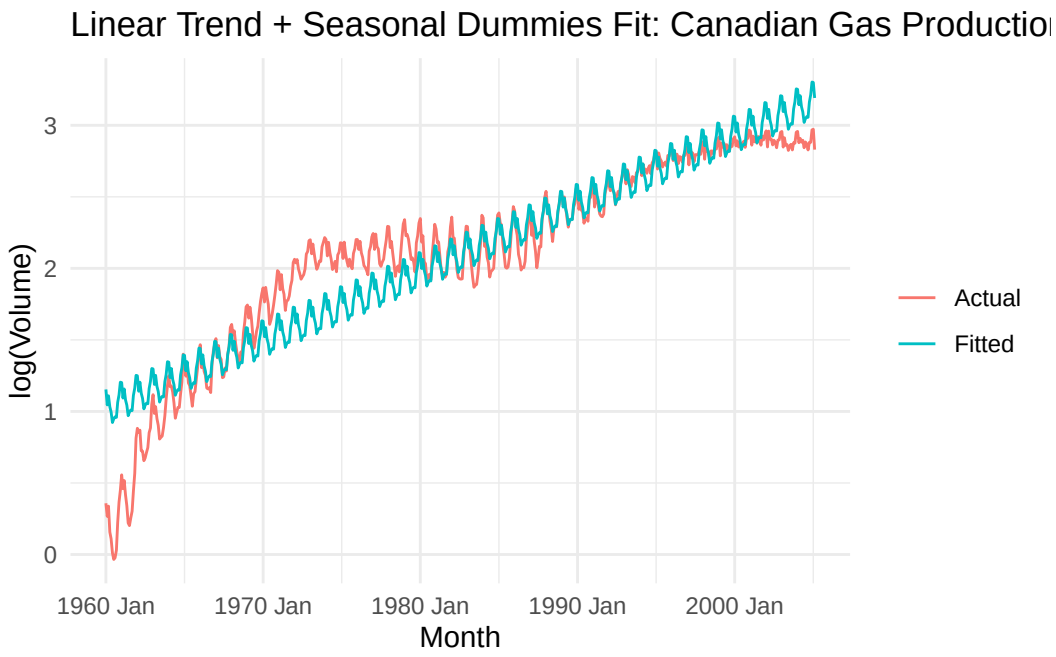
```
can_gas_fit |>
  tidy() |>
  select(term, estimate, std.error, statistic, p.value) |>
  knitr::kable(digits = 3)
```

term	estimate	std.error	statistic	p.value
(Intercept)	1.150	0.041	27.761	0.000
trend()	0.004	0.000	57.692	0.000
season()year2	-0.112	0.052	-2.144	0.032
season()year3	-0.052	0.053	-0.996	0.320
season()year4	-0.133	0.053	-2.537	0.011
season()year5	-0.174	0.053	-3.305	0.001
season()year6	-0.251	0.053	-4.777	0.000
season()year7	-0.230	0.053	-4.374	0.000
season()year8	-0.219	0.053	-4.170	0.000
season()year9	-0.227	0.053	-4.325	0.000
season()year10	-0.125	0.053	-2.383	0.018
season()year11	-0.078	0.053	-1.480	0.140
season()year12	0.007	0.053	0.137	0.891

```
# The estimated average monthly growth rate is .4%. The month with the largest seasonal effe
```

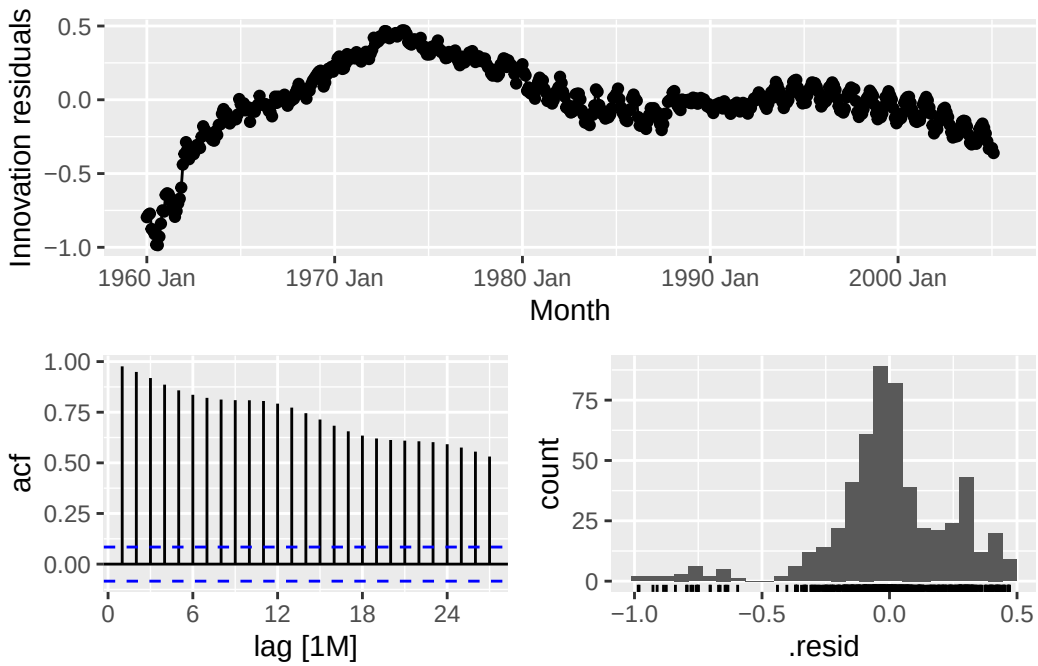
```
can_gas_fit |>
  augment() |>
  ggplot(aes(x = Month)) +
```

```
  geom_line(aes(y = log_volume, color = "Actual")) +
  geom_line(aes(y = .fitted, color = "Fitted")) +
  labs(title = "Linear Trend + Seasonal Dummies Fit: Canadian Gas Production",
       y = "log(Volume)", color = NULL) +
  theme_minimal()
```




```
# The fitted line does not track the actual series very fell. The fitted line is linear, whil
```

```
can_gas_fit |> gg_tsresiduals()
```




```
can_gas_fit |>
  augment() |>
  features(.innov, ljung_box, lag = 24)
```

```
# A tibble: 1 x 3
  .model                                lb_stat lb_pvalue
  <chr>                                   <dbl>     <dbl>
1 TSLM(log_volume ~ trend() + season())   7774.         0
```

```
# The residuals in the graph do not look like white noise; they have a clear trend first goi
```

```
# 5.1 An Expanding Window TSCV
gas_cv_fc <- can_gas |>
  select(Month, Volume) |>
  stretch_tsibble(.init = 240, .step = 1) |>
  model(snaive = SNAIVE(Volume),
        tslm   = TSLM(log(Volume) ~ trend() + season())) |>
  forecast(h = 1)
```

```
gas_cv_fc |>
  accuracy(can_gas, measures = list(MAE = MAE, RMSE = RMSE))
```

```
# A tibble: 2 x 4
  .model .type   MAE  RMSE
  <chr>  <chr> <dbl> <dbl>
1 snaive Test  0.562 0.702
2 tslm   Test  4.01  4.36
```

```
# The date of the first forecast is January 1980, and the date of the last forecast is Februa
```

```
gas_cv_fc |>
  accuracy(can_gas, measures = list(MAE = MAE, RMSE = RMSE)) |>
  select(.model, MAE, RMSE) |>
  knitr::kable(digits = 1)
```

.model	MAE	RMSE
snaive	0.6	0.7
tslm	4.0	4.4

```
# The snaive model forecasts more accurately because the MAE and RMSE are both lower than the
```

```
# 5.2 Is the Difference Significant?
errs <- gas_cv_fc |>
  as_tibble() |>
  left_join(as_tibble(can_gas) |> select(Month, actual = Volume),
            by = "Month") |>
  mutate(e = actual - .mean) |>
  select(Month, .model, e) |>
  pivot_wider(names_from = .model, values_from = e) |>
  na.omit()
```

```
forecast::dm.test(errs$snaive, errs$tslm, h = 1, power = 1)
```

```
    Diebold-Mariano Test

data:  errs$snaiveerrs$tslm
DM = -31.278, Forecast horizon = 1, Loss function power = 1, p-value <
2.2e-16
alternative hypothesis: two.sided
```

```
# DM stat: -31.278, p-value: 2.2e-16. We reject the null that the two model's forecasts are
```

```
# 5.3 A Rolling Window
gas_cv_fc <- can_gas |>
  select(Month, Volume) |>
  slide_tsibble(.size = 240, .step = 1) |>
  model(snaive = SNAIVE(Volume),
        tslm   = TSLM(log(Volume) ~ trend() + season())) |>
  forecast(h = 1)
```

```
gas_cv_fc |>
  accuracy(can_gas, measures = list(MAE = MAE, RMSE = RMSE))
```

```
# A tibble: 2 x 4
  .model .type   MAE  RMSE
  <chr>  <chr> <dbl> <dbl>
1 snaive Test  0.562 0.702
2 tslm   Test  2.09  2.42
```

```
# The tslm model accuracy changed more after the switch to a rolling window. This is because
```

Reflection: The residuals in 4.4 looked like they were missing another differentiation so that they look like white noise. A model that can capture this missing structure looks like a graph that represents a higher degree of differentiation. This would remove the trend of the residuals that is currently happening and make them look like white noise. With the acf graph, the residuals are clearly correlated, and there is a decreasing trend happening in the graph.